Statistical Analysis Report: Income and Sex in the 1994 U.S. Census

Author: Francisco Mateus

Dataset: UCI Adult (Census Income) training extract (32,561 rows, 15 columns)

Project: Conduct a Statistical Analysis Using Python capstone (Adult income analysis)

Date: 28 October 2026

28 October 2026

Overview

I analyzed the UCI Adult (Census Income) dataset (Becker & Kohavi, 1996) to test whether sex is associated with earning over $50K. Of its 32,561 records, 7,841 (24.08%) sit over the $50K line.

This report covers the descriptive statistics, four visual models, and one chi-square test of independence. The result of this analysis points to a clear discrepancy between men and women: men in this data earn over $50K far more often than women, 30.57% against 10.95%. The chi-square test attributes the gap to far more than sampling noise, with a p-value of about 1e-332. The effect size, Cramér's V, is 0.2160, a small-to-moderate association.

Dataset Description

As noted in the Overview, the analysis uses the UCI Adult training extract (Becker & Kohavi, 1996). No earlier project in the program analyzed this dataset. It is a real public extract of the 1994 U.S. Census, hosted by the UCI Machine Learning Repository. Each row is one person, and income is the binary target of whether that person earns over $50K per year.

The table has 15 columns and 32,561 rows: 6 are numeric (age, fnlwgt, education-num, capital-gain, capital-loss, and hours-per-week) and 9 are categorical. The test rests on the categorical pair sex and income, and the descriptive work follows the two numeric variables most relevant to the outcome, age and hours-per-week.

Three columns carry missing values (workclass 1,836, occupation 1,843, and native-country 583, 4,262 cells in total). I keep the missing values in place, because data cleanup is not the purpose of this project. The sex and income columns are complete: all 32,561 rows carry a recorded value for both. The test below therefore uses every row, and the missing cells in the other columns play no part in it.

The load check also flags 24 exact duplicate rows. I keep them, because a census extract can legitimately hold two people with identical recorded attributes. One caution applies further

down: the source is a household survey, and the relationship field lists husband, wife, and own-child entries, so the rows may not be independent.

Methods

Lusa et al. (2024) set out the initial-data-analysis steps that should precede inference: data cleaning, a review of the missing values, and checks on the distribution and shape of each variable. The argument is written for longitudinal studies, but the steps apply to any dataset. Cleanup stays out of scope, which the Dataset Description records, so the screening before any test reduces to two steps: the load check covers the column types, the missing values, and the duplicates, and the descriptive statistics run the shape checks on age and hours-per-week. That screening is the full data preparation, and its results shaped the choices below: a right-skewed age and two unbalanced variables (income and sex).

For the question itself I use a chi-square test of independence. The question asks whether sex and income are associated rather than how much one predicts the other. Both variables are categorical, which is exactly what the test expects. The p-values come from a standard approximation that is only reliable when the expected counts are large, and ours (checked in the Results) sit far above that level.

The null hypothesis (H0) is that sex and income are independent. The alternative (H1) is that they are associated. I fixed the significance level at alpha = 0.05 before the test.

A chi-square test is non-directional: it can say whether an association exists, not which way it runs, so I read the direction of the difference from the observed counts and rates rather than expecting the test to supply them. A second assumption is that every row is a separate observation. That one is not fully true here: the source is a household survey, records can share a household, and the p-value therefore reads only as approximate. The Limitations section takes up that concern.

SciPy's chi2_contingency applies the Yates correction by default to a two-by-two table. I report the uncorrected Pearson statistic (the effect size builds on it) and note the corrected value

in the Results. As an effect size I report Cramér's V, a 0-to-1 measure of the strength of an association (Cramér, 1946). The p-value alone does not show that strength and is easy to misread as the probability that the null hypothesis is true, so the p-value and the effect size travel together (Greenland et al., 2016).

All computations used Python 3.12.14 with pandas 3.0.6, NumPy 2.5.3, and SciPy 1.18.1 (Virtanen et al., 2020), and the figures used Matplotlib 3.11.2 and Seaborn 0.13.2. The notebook runs top to bottom from a clean kernel, writes the four figures itself, and prints every number the report quotes. requirements.txt pins the exact package versions, and the notebook's Reproducibility section documents the whole rebuild with uv step by step, from the environment creation through the executed notebook.

Results

The descriptive statistics show the shape and balance of the four relevant variables:

- age is right-skewed: the mean is 38.5816, the median is 37.0, the standard deviation is 13.6404, the interquartile range is 28.0 to 48.0, and the skewness is 0.5587.
- hours-per-week concentrates at a 40-hour week: the mean is 40.4375, the median is 40.0, the standard deviation is 12.3474, the interquartile range is 40.0 to 45.0, and the skewness is 0.2276.
- income is unbalanced, with 7,841 people (24.08%) over $50K.
- sex is uneven, with 21,790 men and 10,771 women.

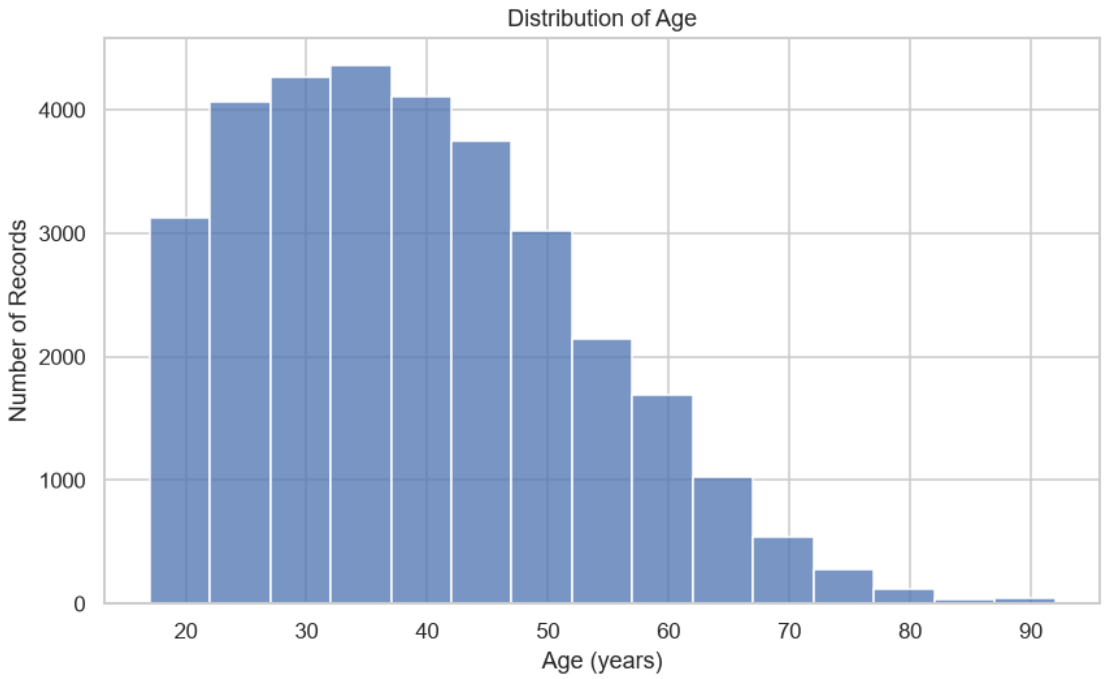


Figure 1. Distribution of age, showing the right skew.

The curve in Figure 1 falls the way the skewness number predicts, but the last populated bin needs its own reading. The notebook's supporting counts show it holds 47 records, with 43 at the 90 cap, 3 at 88, and 1 at 87. The data appear top-coded at 90 (respondents beyond the cap recorded as 90), so that bump reflects the cap rather than a natural peak of older respondents.

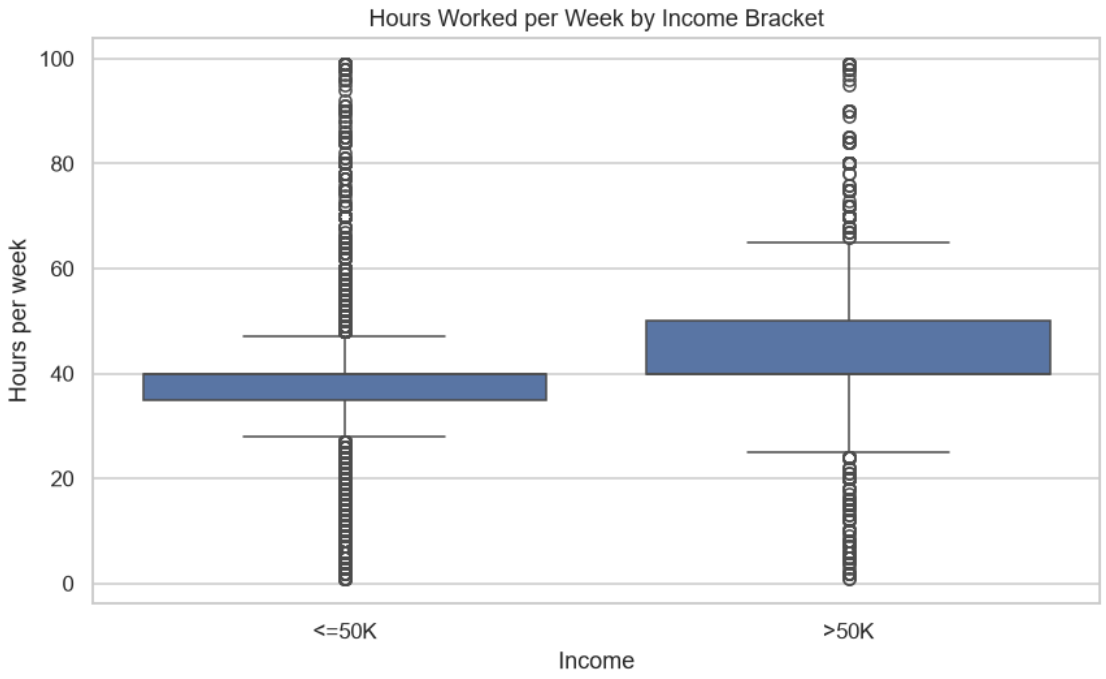


Figure 2. Hours worked per week by income bracket.

Figure 2 shows how hours split across income. Both groups share the median of 40 hours, but the over-$50K group's quartiles sit higher overall (lower quartile 40 against 35, upper quartile 50 against 40), and 49.2% of that group works past 40 hours against 23.2%.

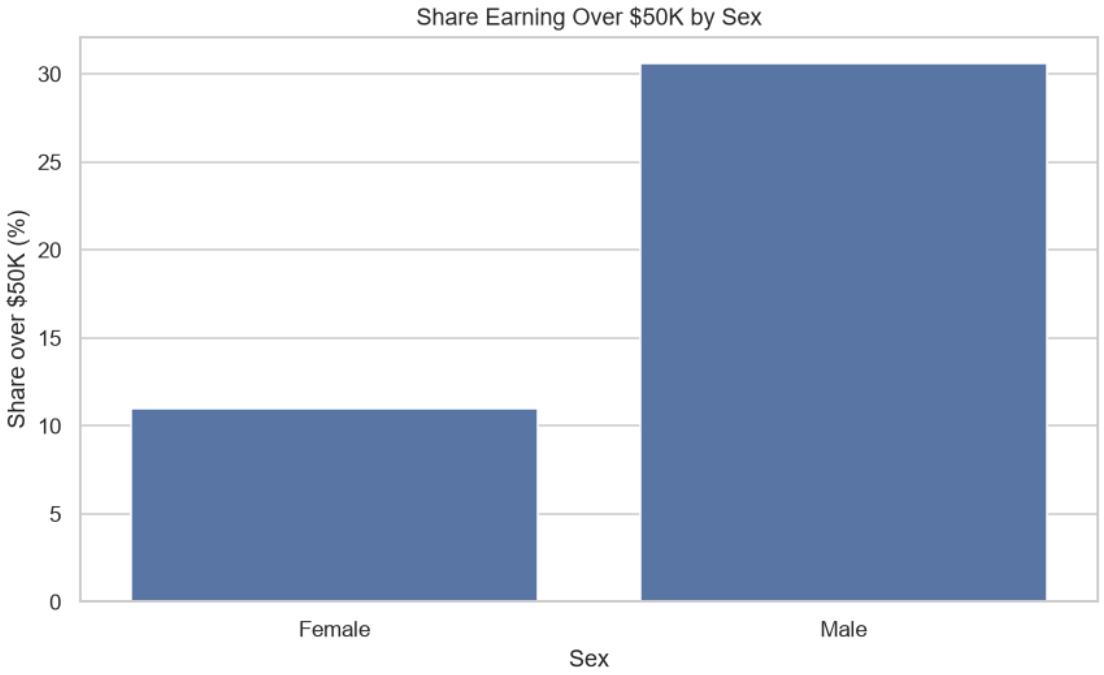


Figure 3. Share earning over $50K by sex.

Figure 3 shows the income rate by sex, the clearest single view of the question. 30.57% of men earn over $50K against 10.95% of women, a gap of 19.63 percentage points.

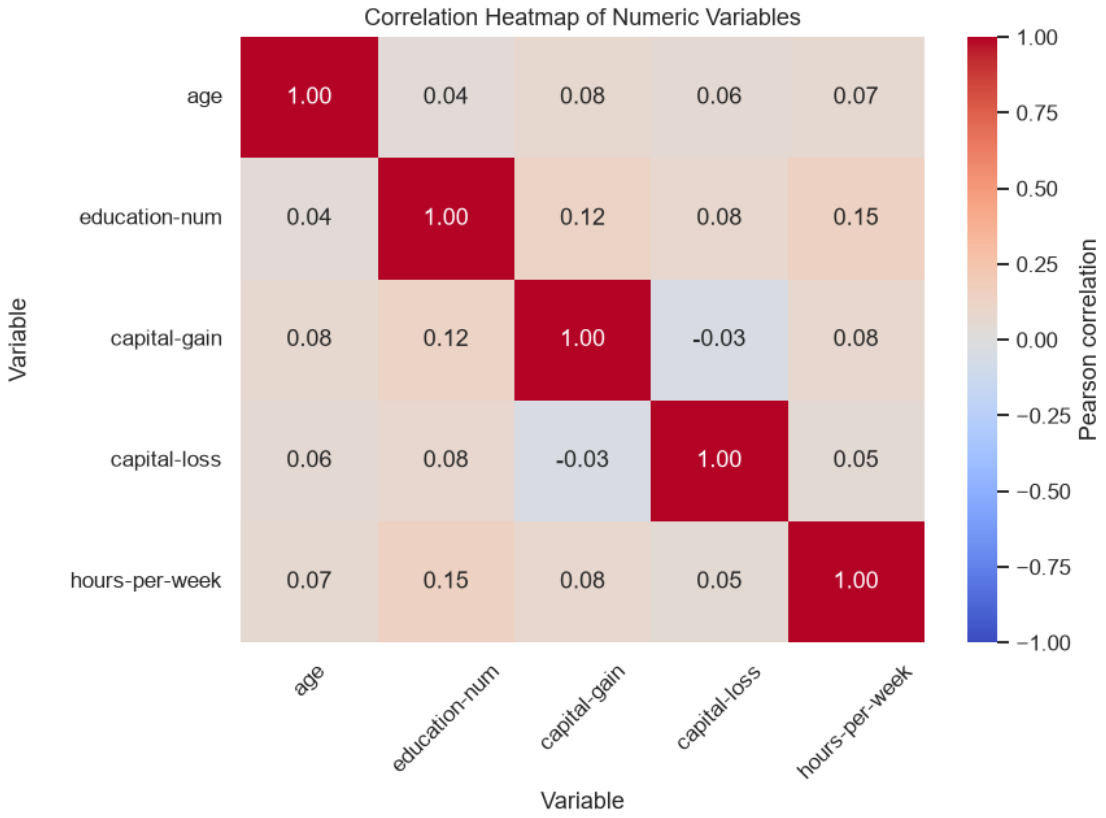


Figure 4. Correlation heatmap of the numeric variables.

The pairwise correlations among the plotted numeric variables are all weak, the largest 0.1481 between education-num and hours-per-week, so the numeric variables do not move together much. That says nothing about their link to income, which is categorical and outside the matrix; Figure 2 shows that at least one numeric variable (hours) tracks income. The matrix also leaves out fnlwgt, a census sampling weight rather than a measured attribute, and carries two heavily zero-inflated columns (91.67% of capital-gain and 95.33% of capital-loss are 0), so those correlations are near-degenerate.

The chi-square test of independence gives a Pearson statistic of 1518.8868 (the Yates-corrected value is 1517.8134) with 1 degree of freedom. The exact p-value underflows double precision, so the notebook reports its order of magnitude, about 1e-332 (below 1e-300).

The smallest expected count is 2593.76, so the approximation is safe. Because the p-value is far below the threshold, I reject the null hypothesis of independence.

The effect size is Cramér's V of 0.2160, computed from the uncorrected Pearson statistic, a small-to-moderate association. For a two-by-two table Cramér's V equals phi, so V squared (0.0466, about 4.7%) is the ratio of the chi-square statistic to the sample size.

The contingency table is:

Sex	**At or below $50K**	**Over $50K**
Female	`9,592`	`1,179`
Male	`15,128`	`6,662`

The gap in the over-$50K rate is 19.63 percentage points, with a 95% confidence interval of [18.78, 20.48] from the normal approximation for a difference in proportions. That interval assumes independent observations, so the household clustering noted in the Methods makes it approximate, exactly as it does the p-value.

Interpretation for a Non-Technical Audience

In this dataset, men were much more likely than women to earn over $50K. About three in ten men were above the threshold, against about one in ten women, a gap of about 19.63 percentage points.

A gap of 19.63 points in a sample of 32,561 would almost never appear by chance if sex and income were unrelated, and the chi-square test confirms the gap (p about 1e-332). The effect size adds the scale: 0.2160 is a small-to-moderate value on the 0-to-1 scale, and sex alone explains only about 4.7% of the variation in who is above the line. Sex is one of many differences between the two groups, not the whole of the difference.

The result is an association, not proof that sex caused the difference in earnings, and it is tied to 1994 and to the United States. The Limitations section lists the other differences between the groups and the other caveats.

Limitations and Potential Bias

The first limitation is that the data are observational. Nobody was assigned to be a man or a woman, so I can report an association between sex and income but not a cause. Unmeasured factors that differ between the groups, such as occupation, education, hours, and access to certain jobs, could contribute to the gap.

The second limitation is the age and scope of the data. The 32,561 records come from a 1994 U.S. Census extract rather than a current or random sample of a defined population, so the gap is a property of this sample and may not generalize. The income target is also a nominal $50K cut in 1994 dollars with no inflation adjustment. sex is coded as a binary Male/Female field, which bounds how far the fairness discussion can go.

The main potential source of bias is the missing and self-reported data. The three columns with missing values (the counts are in the Dataset Description) may be missing for reasons tied to income or sex, in which case the comparison is biased. The kept 24 duplicates and the household clustering behind the relationship field raise the same independence worry: the observations are not all distinct draws, so the p-value and the confidence interval read as approximate.

The result also carries a misuse risk. It describes 1994 census data and says nothing about any individual's ability or potential. Using a demographic attribute such as sex for income, credit, or hiring decisions raises equity and privacy questions, so a difference between group rates is not, on its own, a justification for treating individuals differently.

As a next step, a follow-up analysis could adjust for confounding by modeling income on sex together with education-num, occupation, and hours-per-week. A newer dataset could show how much of the gap persists today.

References

Becker, B., & Kohavi, R. (1996). *Adult* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5XW20

Cramér, H. (1946). *Mathematical methods of statistics* . Princeton University Press.

Greenland, S., Senn, S. J., Rothman, K. J., Carlin, J. B., Poole, C., Goodman, S. N., & Altman, D. G. (2016). Statistical tests, P values, confidence intervals, and power: A guide to misinterpretations. *European Journal of Epidemiology* , 31(4), 337-350. https://doi.org/10.1007/s10654-016-0149-3

Lusa, L., Proust-Lima, C., Schmidt, C. O., Lee, K. J., le Cessie, S., Baillie, M., Lawrence, F., & Huebner, M. (2024). Initial data analysis for longitudinal studies to build a solid foundation for reproducible analysis. *PLOS ONE* , 19(5), e0295726. https://doi.org/10.1371/journal.pone.0295726

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* , 17(3), 261-272. https://doi.org/10.1038/s41592-019-0686-2